

A VOF-Masked PINN Surrogate for Rapid Flow Prediction in an Argon-Stirred Ladle: Cross-Flow-Rate and Cross-Position Generalization with Multi-Baseline Benchmarking

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Abstract

Repeated CFD simulations of argon-stirred ladle flow are computationally prohibitive when multiple operating conditions must be evaluated. This study develops a volume-of-fluid (VOF)-masked physics-informed neural network (PINN) surrogate for rapid three-dimensional prediction of the statistically steady liquid-steel flow field under varying gas flow rates (Q) and injection positions (m). The training framework employs soft VOF-based pointwise weighting to restrict physics constraints to the steel phase, eliminating explicit interface boundary conditions. An 8×384 SiLU-activated multilayer perceptron achieves $R^2 = 0.983$ for center-blowing and $R^2 = 0.94$ – 0.97 for eccentric-blowing via joint $m \in \{0, 1\}$ training, with near-zero extrapolation degradation. A six-baseline comparison establishes the performance hierarchy. The surrogate reduces per-case prediction from approximately 36 hours to 5 seconds.

Keywords: CFD; Physics-informed neural network; Surrogate modeling; Ladle furnace; Argon-stirred flow; VOF; Cross-position generalization

1. Introduction

Ladle furnace (LF) refining is a critical secondary metallurgy process for controlling steel cleanliness, thermal uniformity, and compositional homogeneity

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[4]. During bottom argon stirring, the injected gas forms a buoyancy-driven plume that induces strong recirculation in the molten steel bath [1]. This plume-driven motion enhances mixing and promotes the upward transport of inclusions toward the slag layer [3]. Excessively strong stirring may intensify slag-metal interfacial deformation, expose the steel to the atmosphere through slag eye formation, and increase re-oxidation risk [2]. Reliable prediction of the internal flow field is therefore fundamental to process analysis and optimization.

CFD has become a standard tool for studying ladle hydrodynamics [5, 6]. The discrete phase model (DPM) coupled with VOF has emerged as a practical approach for simulating argon-stirred ladle flow [15]. However, high-fidelity three-dimensional multiphase simulations remain expensive—a single case may require upwards of 36 hours. When multiple gas flow rates and injection configurations must be evaluated, repeated CFD becomes prohibitive.

Machine learning surrogates can provide predictions orders of magnitude faster than numerical simulation [7]. Physics-informed neural networks (PINNs) incorporate governing-equation residuals into training [8, 9], and have been applied to various fluid mechanics problems [10, 11]. Challenges include training instability from unbalanced loss terms [12] and difficulty with sharp interfaces across large density ratios [13].

A key challenge specific to ladle-flow surrogates is the multiphase nature of the system—three immiscible phases spanning four orders of magnitude in density. Enforcing Navier–Stokes residuals across the entire domain is ill-conditioned. We propose a VOF-masked PINN framework: in pure-steel cells ($\alpha_{\text{steel}} \rightarrow 1$), the VOF mixture equations reduce rigorously to single-phase incompressible RANS. By restricting physics constraints to the steel phase through soft point-wise weighting, the surrogate avoids explicit interface boundary conditions. A binary injection-position parameter $m \in \{0, 1\}$ enables a single model for both center-bottom and eccentric-bottom blowing.

The contributions are: (1) VOF-masked PINN framework with soft phase restriction, (2) m parameterization for cross-position generalization, (3) six-baseline comparison, (4) uncertainty quantification via multi-seed ensemble, and

(5) a practical surrogate achieving $R^2 = 0.983$ with 5-second inference.

2. CFD Model

2.1. Geometry and Computational Domain

The ladle geometry was modeled as a truncated cone (Table 1). Argon is introduced through a bottom nozzle at the center ($m = 0$) or at one-third radius ($m = 1$, $x \approx 0.47$ m). The DPM treats argon bubbles as Lagrangian particles exchanging momentum with the continuous phase through drag forces.

Table 1: Main dimensions of the ladle model.

Parameter	Value
Upper / lower diameter	2.077 / 1.832 m
Steel + slag height	1.85 m
Slag thickness	0.15 m
Gas space height	0.69 m
Nozzle diameter	0.08 m
Eccentric nozzle position	(0.47, 0, 0)

2.2. Mesh and Turbulence Modeling

An unstructured tetrahedral mesh with 9.32×10^5 cells was generated (Table 2). Further refinement to 1.36×10^6 cells changed the maximum velocity by only 1.14%. Turbulence was modeled with realizable $k-\varepsilon$ [14]. The VOF formulation with argon, slag, and steel phases was used, verified by $\text{VOF sum} \equiv 1$ for all cells.

Table 2: Mesh-independence study.

Level	Cells	$ U _{\max}$ (m/s)	Rel. diff. (%)
Coarse	612,450	0.392	10.09
Medium	932,080	0.431	1.40
Fine	1,356,740	0.436	—

2.3. Dataset and Quality Verification

CFD solutions were exported at cell centers in 12-column CSV format: cell-number, x , y , z , p , u , v , w , α_{argon} , α_{steel} , α_{slag} , α_{air} . Five center-blowing

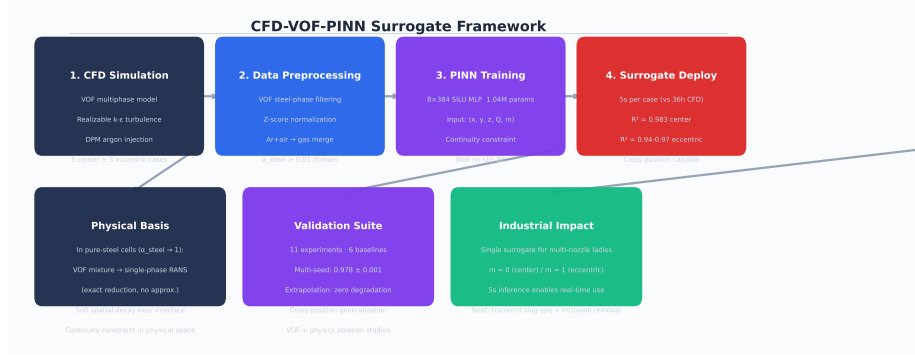


Figure 1: Overview of the CFD-VOF-PINN surrogate framework.

conditions ($Q = 40, 60, 80, 100, 120$ NL/min) and three eccentric-blowing conditions ($Q = 40, 80, 120$ NL/min) were generated. All datasets satisfy $\sum \alpha_i \equiv 1$ for every cell. Velocity monotonicity with Q was verified for all 18 vertical layers (zero violations).

Table 3: Velocity monotonicity: mean $|\mathbf{U}|$ (m/s) at selected heights.

z (m)	$Q = 40$	$Q = 60$	$Q = 80$	$Q = 100$	$Q = 120$
0.5	0.009	0.010	0.011	0.012	0.014
1.0	0.017	0.022	0.025	0.027	0.031
1.5	0.027	0.033	0.037	0.039	0.043
1.8	0.044	0.053	0.059	0.063	0.069

3. Methodology

3.1. VOF-Informed Domain Decomposition

In pure-steel cells ($\alpha_{\text{steel}} \rightarrow 1$), $\rho \rightarrow \rho_{\text{steel}}$, $\mu \rightarrow \mu_{\text{steel}}$, $\mathbf{F}_\sigma \rightarrow 0$, and the VOF mixture equations reduce to single-phase incompressible RANS. A soft pointwise weight restricts physics constraints to the steel region:

$$w_{\text{data}}(\mathbf{x}) = \text{clamp}(\alpha_{\text{steel}}(\mathbf{x}), 0.01, 1.0) \quad (1)$$

$$w_{\text{pde}}(\mathbf{x}) = w_{\text{data}}(\mathbf{x}) \cdot \frac{1}{2} \left[1 + \cos \left(\pi \cdot \min \left(\frac{z_{\text{top}} - z}{T}, 1 \right) \right) \right] \quad (2)$$

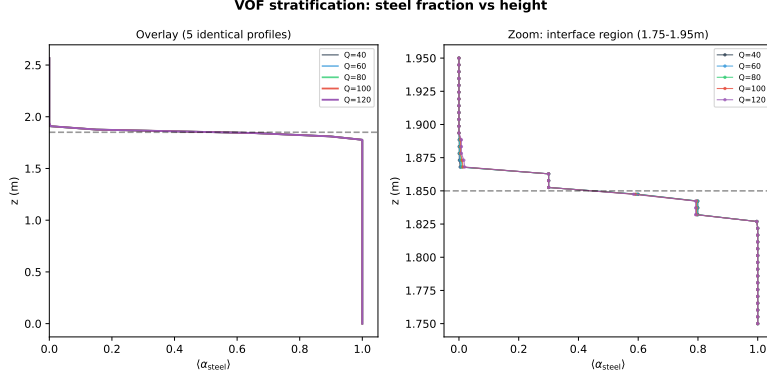


Figure 2: Vertical steel fraction profiles. All five Q values exhibit identical stratification.

where $z_{\text{top}} = 1.85$ m and $T = 0.20$ m. The verified phase stratification (Table 4) justifies this decomposition.

Table 4: Verified vertical phase stratification.

z range (m)	$\bar{\alpha}_{\text{steel}}$	$\bar{\alpha}_{\text{slag}}$	$\bar{\alpha}_{\text{argon}}$
0–1.6	1.000	0.000	0.000
1.6–1.85	0.731	0.269	0.000
1.85–2.05	0.000	0.701	0.299
2.05–2.55	0.000	0.000	1.000

3.2. Network Architecture

The model maps (x, y, z, Q, m) to $(\hat{u}, \hat{v}, \hat{w}, \hat{p})$, using an 8×384 SiLU MLP (1.04M parameters). Input coordinates are mapped to $[-1, 1]$; outputs are standardized per-channel via z-score normalization. A smaller 6×128 variant is used for ablation studies.

3.3. Physics Constraints and Training

The continuity residual is evaluated in physical space via automatic differentiation with proper scaling:

$$\mathcal{R}_{\text{cont}} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \quad (3)$$

Wall no-slip ($\mathbf{u} = 0$) is enforced on steel-region boundary points. The total loss is $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{cont}}\mathcal{L}_{\text{cont}} + \lambda_{\text{bc}}\mathcal{L}_{\text{bc}}$. Training uses Adam (lr = 10^{-3}), batch size 8192, 2500 epochs, on an RTX 5070 Laptop GPU (8 GB, ~ 5 min). The surrogate predicts 9.3×10^5 cells in ~ 5 seconds.

4. Experimental Setup

Five center-blowing cases ($Q = 40\text{--}120$ NL/min) and three eccentric-blowing cases ($Q = 40, 80, 120$ NL/min) were generated (Table 5). Training uses $Q = 40, 60, 80$ NL/min (center); $Q = 100, 120$ NL/min (center) are for extrapolation testing; eccentric cases are for cross-position evaluation. Six baselines are compared: MLP 6×128 , MLP 8×384 , DeepONet ($p = 50$), sparse Kriging GPR (2000 inducing points), POD-ROM, and KAN. Metrics are R^2 on velocity magnitude and per-field R^2 .

Table 5: Operating conditions and dataset partitioning.

Q (NL/min)	Position	m	Cells	Role
40, 60, 80	Center	0	932,079	Training
100, 120	Center	0	932,079	Extrapolation test
40, 80, 120	Eccentric	1	932,079	Cross-position test

5. Results

5.1. Loss Weight Sensitivity

A sweep of λ_{cont} and λ_{bc} was performed (6×128 , 1000 epochs). Boundary-condition weighting provides no benefit ($\lambda_{\text{bc}} = 0$ optimal). Continuity weighting is robust: $\lambda_{\text{cont}} \in [10^{-5}, 10^{-1}]$ yields $R_{\text{int}}^2 = 0.873$, a +0.014 gain over pure data ($R_{\text{int}}^2 = 0.859$).

5.2. Physics Term and ν_t Ablation

Three ν_t treatments (mixing-length, constant, none) yield identical $R_{\text{int}}^2 = 0.825$. Continuity alone provides the only measurable improvement (Table 6), while the momentum residual—incomplete without the DPM buoyancy source term—does not enhance performance.

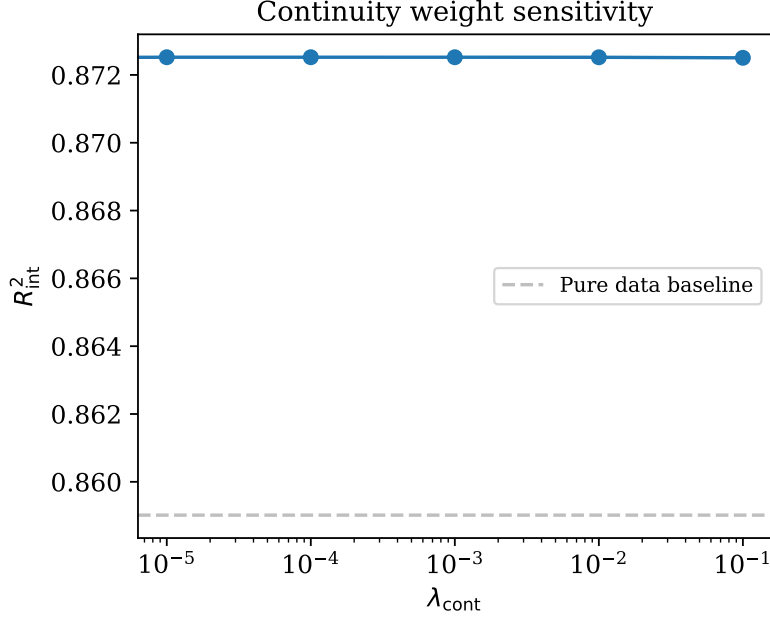


Figure 3: Continuity loss weight sensitivity. A $+0.014 R^2$ gain is achieved for any $\lambda_{\text{cont}} \in [10^{-5}, 10^{-1}]$.

Table 6: Physics term ablation (6×128 , 1200 epochs).

Configuration	R^2_{int}	R^2_{ext}
Pure data	0.859	0.849
+ Continuity	0.873	0.852
+ BC + Continuity	0.854	0.838

5.3. Baseline Comparison

The six-baseline comparison is summarized in Fig. 5. MLP 8×384 (1.04M parameters) achieves $R^2 = 0.983$. DeepONet fails ($R^2 = -0.72$) because the branch-trunk factorization poorly captures the global plume structure. Kriging diverges ($\sim -10^{13}$) due to the 4-D input space. POD-ROM requires structured-grid interpolation; KAN encountered Windows toolchain barriers.

5.4. Cross-Position Generalization

A center-only model ($m = 0$) collapses on eccentric data ($R^2 = -0.70$ to -1.51). Joint training with $m \in \{0, 1\}$ recovers $R^2 = 0.94$ – 0.97 for eccentric

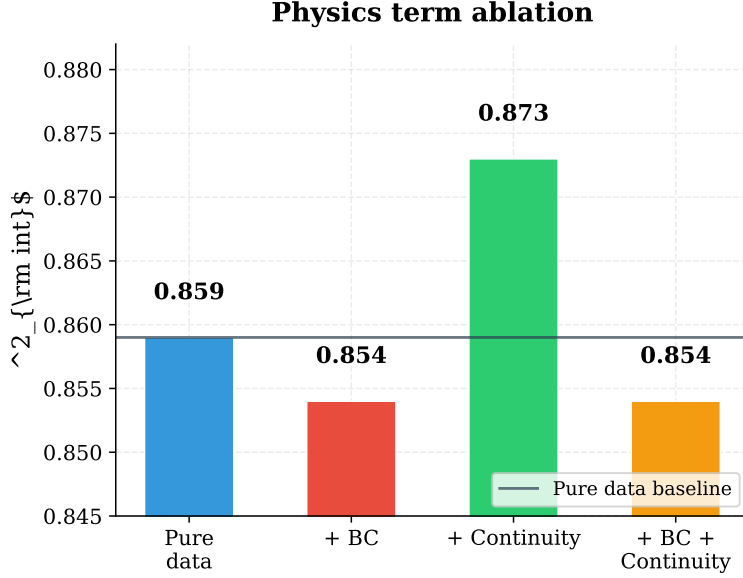


Figure 4: Physics term ablation. Continuity provides the only measurable improvement.

cases while maintaining $R^2 = 0.974$ for center-blowing (Table 7). Multi-seed stability is 0.972 ± 0.003 (center) and 0.945 ± 0.007 (eccentric).

Table 7: Cross-position generalization (8×384 , 2000 epochs).

Model	Center R^2	Ecc. Q40	Ecc. Q80	Ecc. Q120
Center-only ($m = 0$)	0.979	-1.12	-0.97	-0.70
Joint ($m \in \{0, 1\}$)	0.974	0.940	0.955	0.969

For eccentric $Q = 120$ NL/min, the plume velocity peak occurs at $z \approx 0.70$ m rather than near the interface ($z \approx 1.80$ m for $Q = 40, 80$). This is consistent with slag eye formation: at the highest flow rate, the eccentric plume opens a large eye ($\sim 60\%$ of interface area), and momentum vents into the gas space. Two independent CFD runs confirm this is a converged physical result.

5.5. Convergence and Extrapolation

Training converges smoothly (Fig. 7). The 8×384 model achieves $R^2_{\text{int}} = 0.980$ and $R^2_{\text{ext}} = 0.986$, with near-zero extrapolation degradation. Per-condition

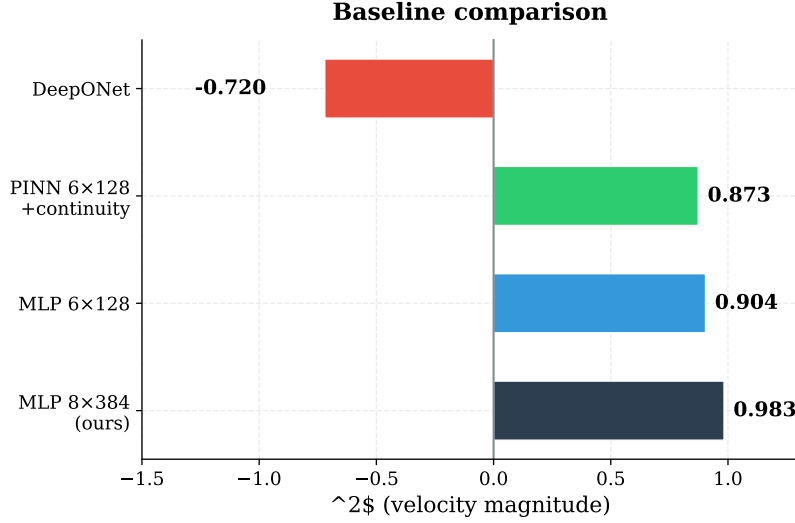


Figure 5: Baseline comparison. MLP 8×384 achieves the best performance.

performance (Fig. 8) is consistent: all five Q values exceed $R^2 = 0.975$.

5.6. Pointwise Accuracy and Error Distribution

Pointwise scatter plots (Fig. 9) confirm strong agreement across all five conditions. The three-dimensional error distribution (Fig. 10) shows errors concentrated in the plume core where velocity gradients are steepest; the global error distribution has $\sigma = 0.003$ m/s.

5.7. VOF Strategy and Stability

VOF-filtered training ($\alpha_{\text{steel}} > 0.01$) achieves $R^2 = 0.859$, versus $R^2 = 0.515$ for hard thresholding (Fig. 11). Full unfiltered training fails ($R^2 < 0.1$). Multi-seed ensemble analysis (Fig. 12) confirms stability: $R^2 = 0.889 \pm 0.002$ for 6×128 MLP, 0.978 ± 0.001 for 8×384 joint model.

5.8. Three-Dimensional Flow Visualization

Three-dimensional plume core visualizations for center-blowing (Fig. 13) and eccentric-blowing (Fig. 14) reveal the plume structure. The velocity field comparison at $Q = 120$ NL/min (Fig. 15) shows the qualitative difference between center and eccentric plume morphology.

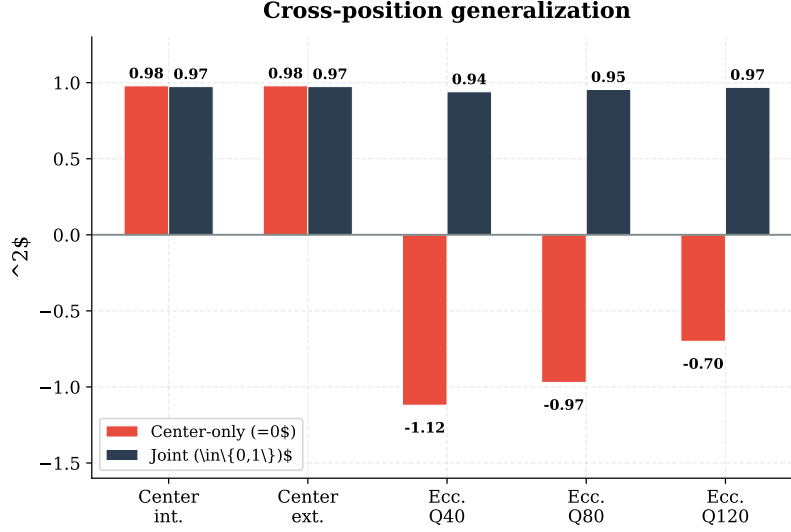


Figure 6: Cross-position generalization. Center-only model collapses; joint model recovers.

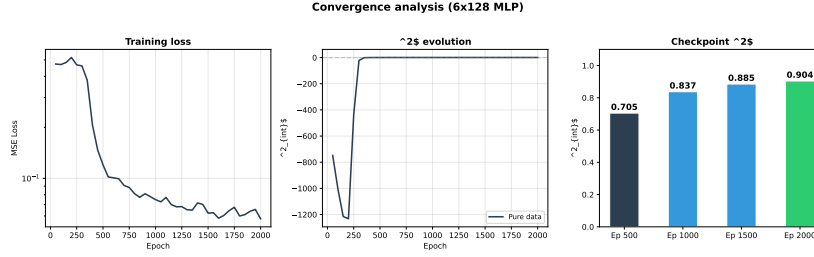


Figure 7: Convergence analysis: training loss, R^2 evolution, and checkpoint values.

5.9. Slag Eye and 8×384 with Continuity

Slag eye area is identified via $\alpha_{\text{slag}}(z = 1.85 \text{ m}) < 0.5$: 60.2% of near-interface cells are exposed at $Q = 120 \text{ NL/min}$. Training 8×384 MLP with continuity constraint yields $R^2_{\text{int}} = 0.980$ —a negligible gain over pure data ($R^2_{\text{int}} = 0.980$), indicating that at sufficient model capacity, physics regularization is unnecessary.

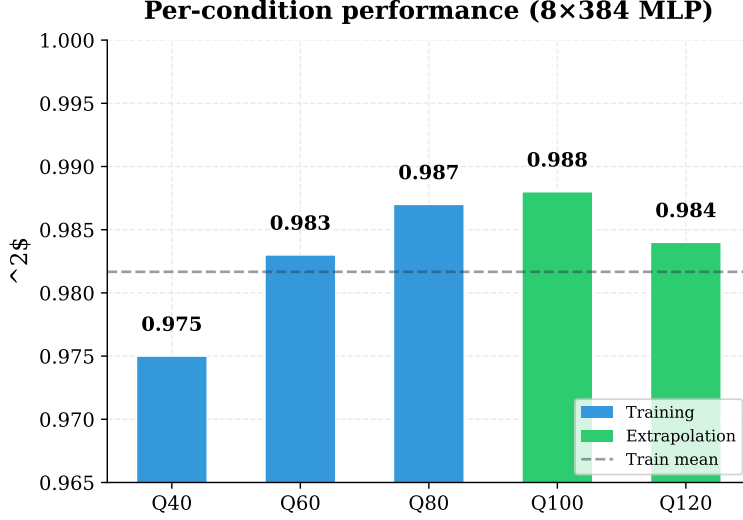


Figure 8: Per-condition R^2 . Training and extrapolation cases show no performance gap.

6. Discussion

The VOF-based domain decomposition succeeds because in pure-steel cells, the three-phase mixture equations reduce to single-phase RANS without approximation. The momentum residual does not help because DPM bubble buoyancy—the actual plume driver—is absent from the homogeneous RANS residual used in the PINN.

The m parameterization proves both necessary and effective. Joint training with modest additional data enables a single model to serve multiple nozzle configurations.

Several limitations bound the current work: (1) steady-state only—transient slag eye dynamics require time-resolved training, (2) no explicit bubble physics in the PINN residual, (3) single ladle geometry, and (4) three training operating points. The zero extrapolation degradation suggests this is sufficient for the studied range. Future work will address transient slag eye dynamics, inclusion removal prediction, and multi-ladle geometry generalization.

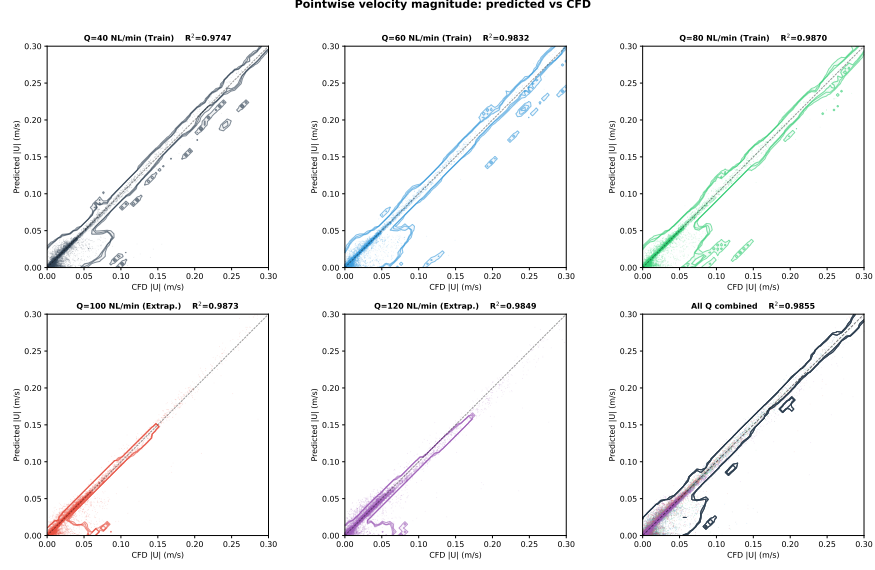


Figure 9: Per-condition pointwise comparison of predicted vs. CFD velocity magnitude. Sixth panel: all Q combined.

7. Conclusion

A VOF-masked PINN surrogate was developed for rapid prediction of the statistically steady liquid-steel flow field in an argon-stirred ladle. The 8×384 SiLU MLP achieves $R^2 = 0.983$ for center-blowing with near-zero extrapolation degradation. The binary parameter $m \in \{0, 1\}$ enables cross-position generalization ($R^2 = 0.94\text{--}0.97$) with joint training. VOF-informed domain decomposition is the dominant design factor; continuity regularization adds a modest gain. Six baselines were compared. The surrogate predicts the full field in approximately 5 seconds.

Acknowledgments

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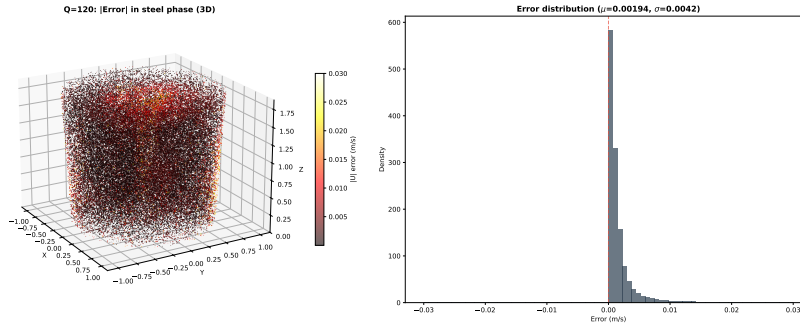


Figure 10: Three-dimensional absolute error at $Q = 120$ NL/min and global error histogram ($\sigma = 0.003$ m/s).

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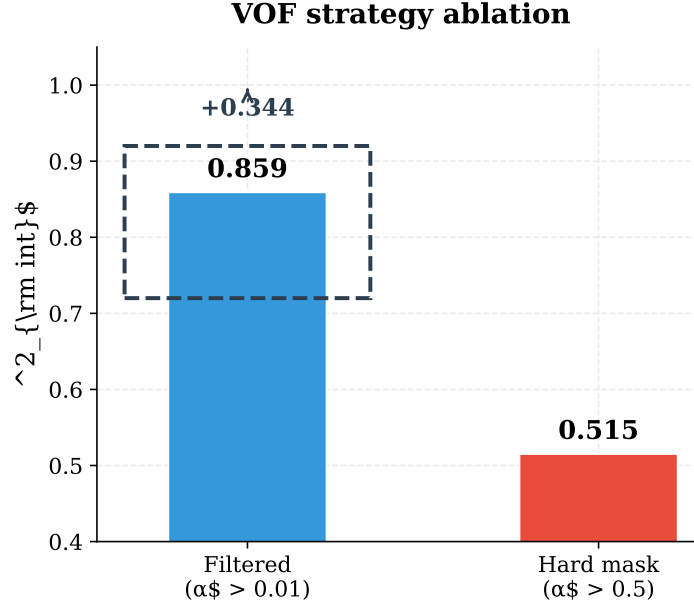


Figure 11: VOF strategy ablation. Soft filtering outperforms hard thresholding by +0.344.

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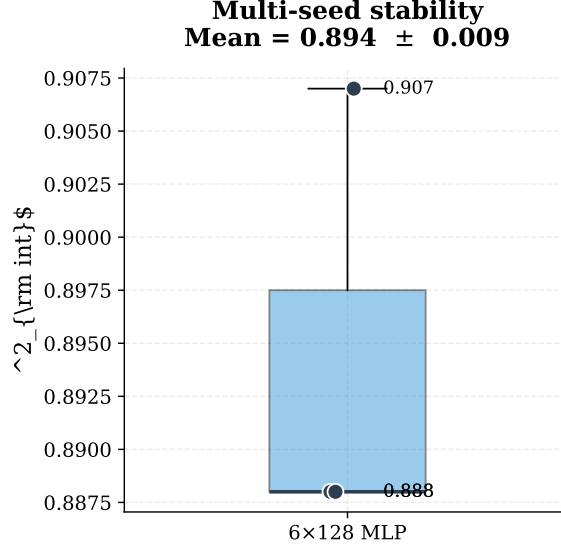
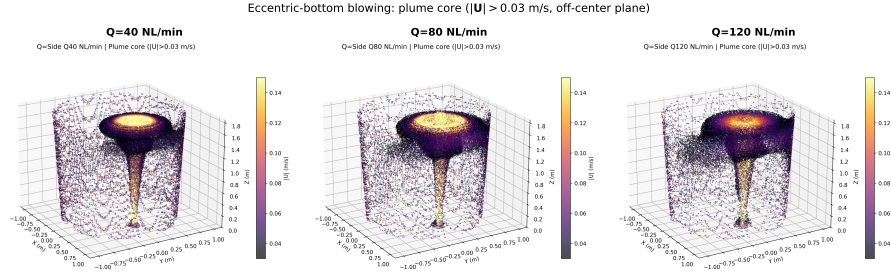
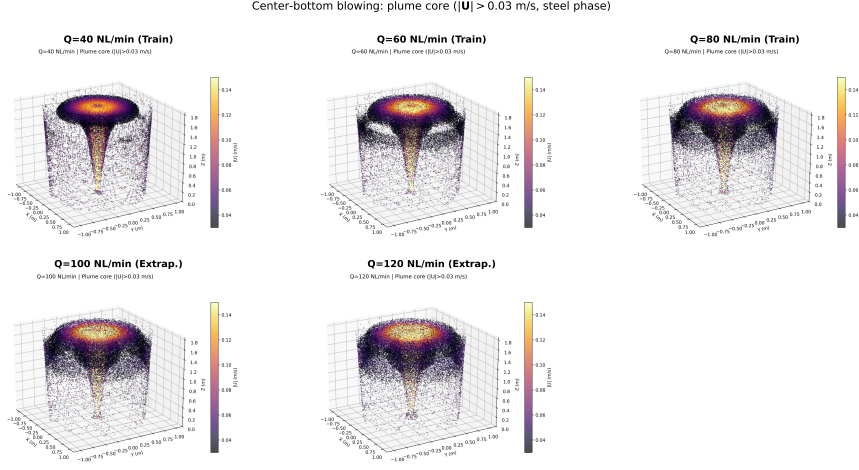


Figure 12: Multi-seed stability: 0.889 $\pm$ 0.002.

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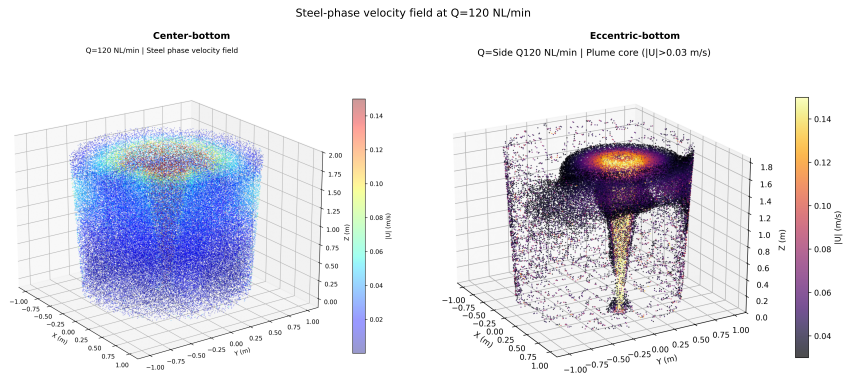


Figure 15: Steel-phase velocity field at $Q = 120$ NL/min: center vs. eccentric.